Aide-mémoire des commandes utiles en TP

1 Configurations initiales

```
Router> enable
Router# configure terminal
Router(config)# hostname R1                              ! Assign device name
R1(config)# enable secret cisco                          ! Encrypt Privileged EXEC password
R1(config)# banner motd # Authorized Access Only! #      ! Set MOTD warning banner
R1(config)# service password-encryption                  ! Encrypt cleartext passwords
```

2 Sécurisation des accès (Console, VTY, & SSH)

2.1 Sécurité basique

```
R1(config)# line console 0
R1(config-line)# password cisco                          ! Set console password
R1(config-line)# login                                   ! Enable password check
R1(config-line)# logging synchronous                     ! Prevent syslog interruption
R1(config-line)# exit

R1(config)# line vty 0 15
R1(config-line)# password cisco                          ! Set Telnet/SSH line password
R1(config-line)# login
R1(config-line)# exit
```

2.2 Sécurité d'accès à distance par ssh

```
R1(config)# ip domain-name example.com                   ! Required for RSA key generation
R1(config)# crypto key generate rsa general-keys modulus 2048   ! Generate encryption key
R1(config)# username admin secret AdminPass123           ! Create local admin user
R1(config)# line vty 0 15
R1(config-line)# transport input ssh                     ! Restrict VTY lines to SSH
R1(config-line)# login local                             ! Authenticate via local DB
R1(config-line)# exit
```

2.3 Sécurité plus avancée

```
S1(config)# login block-for 120 attempts 3 within 60     ! Prevents brute-force password attacks
S1(config)# line vty 0 15
S1(config-line)# exec-timeout 20                         ! Set an inactivy timer
S1(config)# interface range gigabitethernet 0/1 - 5
S1(config-line)# no shutdown                             ! Desactivate a range of interfaces
```

3 Configuration des interfaces et des adresses

3.1 Interfaces des routeurs

```
R1(config)# ipv6 unicast-routing                          ! Enable IPv6 routing (Routers)
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# description Link to LAN-A
R1(config-if)# ip address 192.168.1.1 255.255.255.0      ! Assign IPv4 address & mask
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64         ! Assign IPv6 GUA
R1(config-if)# ipv6 address fe80::1 link-local            ! Assign IPv6 link-local address
R1(config-if)# no shutdown                                ! Enable the interface
```

3.2 Interface de gestion des commutateurs

```
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.2 255.255.255.0      ! Assign IPv4 management address
S1(config-if)# no shutdown                                ! Enable virtual interface
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.1.1                ! Set default gateway
```

4 Vérification et dépannage

```
S1# show running-config                                   ! Active config in RAM
S1# show startup-config                                   ! Saved config in NVRAM
S1# show ip interface brief                               ! IPv4 interface status summary
S1# show ipv6 interface brief                             ! IPv6 interface status summary
S1# show interfaces [interface-id]                        ! Detailed stats (duplex, MAC, etc.)
S1# show ip route                                         ! IPv4 routing table
S1# show ipv6 route                                       ! IPv6 routing table
S1# show ip arp                                  ! Display the full IP-to-MAC resolution table
S1# ping <ip-address>                                     ! Test Layer 3 reachability
S1# traceroute <ip-address>                               ! Trace path to destination
```

5 Gestion des configurations et opérations sur les fichiers

```
S1# copy running-config startup-config                    ! Save RAM config to NVRAM
S1# erase startup-config                                  ! Clear configuration in NVRAM
S1# delete vlan.dat                                       ! Delete VLAN database (Switches)
S1# reload                                                ! Reboot device
```

6 Commandes de configuration sous Windows

```
C:\> ipconfig                                             ! Display basic IP configuration
C:\> ipconfig /all                                        ! Display IP, MAC, DNS, DHCP details
C:\> ping <ip-address>                                    ! Test connectivity
C:\> tracert <ip-address>                                 ! Trace route to target
C:\> nslookup <domain-name>                               ! Query DNS server details
```